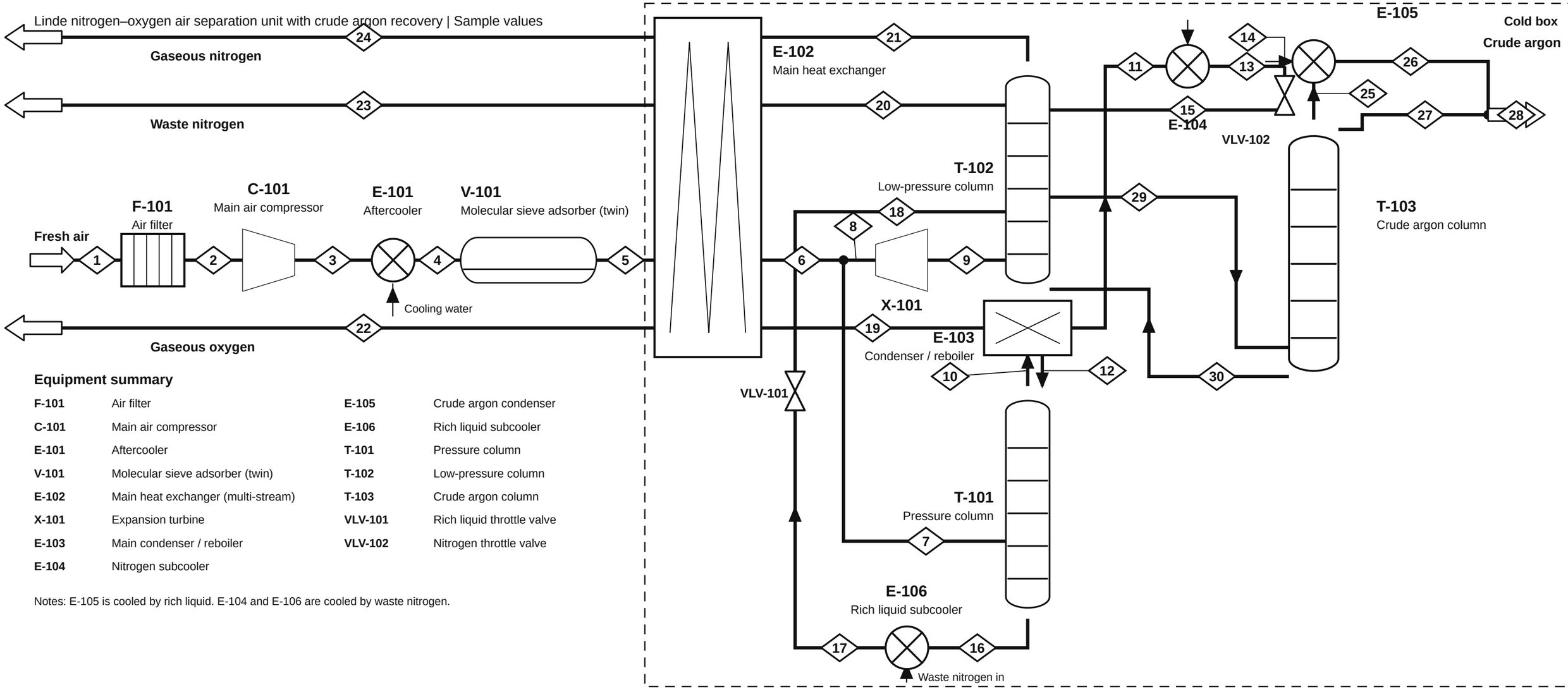
PROCESS FLOW DIAGRAM

Linde nitrogen–oxygen air separation unit with crude argon recovery | Sample values



Equipment summary

F-101	Air filter	**E-105**	Crude argon condenser
C-101	Main air compressor	**E-106**	Rich liquid subcooler
E-101	Aftercooler	**T-101**	Pressure column
V-101	Molecular sieve adsorber (twin)	**T-102**	Low-pressure column
E-102	Main heat exchanger (multi-stream)	**T-103**	Crude argon column
X-101	Expansion turbine	**VLV-101**	Rich liquid throttle valve
E-103	Main condenser / reboiler	**VLV-102**	Nitrogen throttle valve
E-104	Nitrogen subcooler		

Notes: E-105 is cooled by rich liquid. E-104 and E-106 are cooled by waste nitrogen.

Stream table — flow rates, temperatures, pressures and compositions are sample values

Stream	Flow rate (kmol/h)	Temperature (°C)	Pressure (kPa)	Nitrogen (mol %)	Oxygen (mol %)	Argon (mol %)	Phase
1	1000	20	101	78.0	21.0	1.0	Vapour
2	1000	20	100	78.0	21.0	1.0	Vapour
3	1000	175	625	78.0	21.0	1.0	Vapour
4	1000	22	600	78.0	21.0	1.0	Vapour
5	1000	20	590	78.0	21.0	1.0	Vapour
6	1000	−171	568	78.0	21.0	1.0	Vapour
7	800	−171	568	78.0	21.0	1.0	Vapour
8	200	−171	568	78.0	21.0	1.0	Vapour
9	200	−182	138	78.0	21.0	1.0	Vapour
10	600	−178	560	97.0	2.0	1.0	Vapour
11	400	−178	555	97.0	2.0	1.0	Liquid
12	200	−178	550	97.0	2.0	1.0	Liquid
13	400	−186	545	97.0	2.0	1.0	Liquid
14	400	−186	540	97.0	2.0	1.0	Liquid
15	400	−192	130	97.0	2.0	1.0	Liquid

Stream	Flow rate (kmol/h)	Temperature (°C)	Pressure (kPa)	Nitrogen (mol %)	Oxygen (mol %)	Argon (mol %)	Phase
16	400	−174	558	59.0	40.0	1.0	Liquid
17	400	−181	552	59.0	40.0	1.0	Liquid
18	400	−184	138	59.0	40.0	1.0	Two-phase
19	200	−179	130	1.0	99.0	0.0	Vapour
20	290	−185	125	96.6	3.1	0.3	Vapour
21	500	−192	120	99.5	0.5	0.0	Vapour
22	200	20	115	1.0	99.0	0.0	Vapour
23	290	20	105	96.6	3.1	0.3	Vapour
24	500	20	110	99.5	0.5	0.0	Vapour
25	30	−184	124	5.0	5.0	90.0	Vapour
26	30	−184	120	5.0	5.0	90.0	Liquid
27	20	−184	120	5.0	5.0	90.0	Liquid
28	10	−184	120	5.0	5.0	90.0	Liquid
29	100	−172	130	1.0	90.0	9.0	Vapour
30	90	−176	128	0.6	99.4	0.0	Liquid